Spatial Hypertext as a Mixed-Initiative Interface for Enhancing Critical Engagement on Social Media

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Abstract

Modern social media often prioritizes engagement metrics over user reflection, leading to shallow interaction and reduced critical thinking. This concept paper proposes a theoretical design framework combining spatial hypertext with mixed-initiative interaction to support user-led exploration and reflection on digital content. The design emphasizes human-computer collaboration, transparency, and user agency. Grounded in HCI ethics, the paper draws from established research in hypertext theory, interface design, and digital literacy.

Keywords: Spatial hypertext, mixed-initiative systems, human-computer interaction, social media, digital literacy, critical reflection

Introduction and Related Work

Social media interfaces today are optimized for speed, virality, and engagement, often at the cost of thoughtful interaction [1]. Studies highlight how such platforms contribute to echo chambers, filter bubbles, and reduced user autonomy [2].

To address these challenges, this concept builds on spatial hypertext theory [3], which supports meaning-making through visual arrangement rather than linear links. It integrates mixed-initiative principles from Horvitz [4], where the system collaborates with the user by offering help without taking control.

In HCI, researchers such as Shneiderman [5] and Amershi et al. [6] emphasize the importance of user agency, transparency, and control when designing intelligent systems. Atzenbeck et al. [7] frame spatial hypertext as a medium for cognitive mapping that enables the synthesis of automation and user-driven augmentation in intelligent systems.

These works collectively support the creation of media interfaces that promote critical engagement and reflective interaction.

Conceptual Framework

The proposed interface presents social media posts as movable, annotatable tiles on a 2D canvas. Users manually group and annotate content by topic, bias, tone, or credibility, supporting flexible meaning-making in exploratory and emotionally driven tasks [3, 8]. As demonstrated in personal information interfaces, such externalization of thought facilitates deeper cognitive engagement and reflection over passive consumption [9].

Unlike algorithmic feeds that impose linear, system-driven priorities [3,8], this design observes user-created structures and offers context-aware suggestions, such as related sources, contrasting viewpoints, or overlooked perspectives. These interventions adapt to user behavior without interrupting flow or overriding control, aligning with Horvitz's principles of mixed-initiative interaction [4], Amershi et al.'s findings on maintaining user intent in intelligent systems [6], and recent applications of spatial hypertext in recommender systems that support user-driven structuring and contextual suggestions without disrupting control [10].

This conceptual structure integrates three key foundations:

- **Spatial hypertext:** Meaning emerges through layout, spatial proximity, and annotations rather than through fixed link hierarchies [3].
- **Mixed-initiative systems:** The system collaborates by offering contextual help, guided by confidence thresholds, while preserving user autonomy [4,6].
- **Human-centered ethics:** Interfaces are designed to promote transparency, trust, and agency, ensuring that augmentation serves, rather than replaces, human judgment [5,7].

Use Cases and Evaluation Directions

Use Case 1: Civic Reflection. A user clusters articles and tweets around a political topic to identify echo chambers and compare perspectives [1,2].

Use Case 2: Personal Sensemaking. Users group and annotate their social media content history to reflect on emotional tone or topic evolution [3,8].

Use Case 3: Classroom Mapping. Educators use the system to structure debates and collaboratively visualize diverse student viewpoints and arguments around controversial topics [7].

Evaluation Themes (from literature):

- *Cognitive engagement:* Metrics of how spatial arrangement supports bias awareness, measured by time spent arranging, depth of annotations, and layout complexity [3]
- *User-system interaction:* Reliability and perceived usefulness of system suggestions [6]
- *Ethical reflection:* Evidence of increased critical awareness in interaction, analysis of user-generated layouts and annotations, looking for evidence of perspective broadening [5]

Discussion and Conclusion

This concept reimagines social media as a reflective, user-directed space, grounded in spatial hypertext and HCI principles. Its structure is informed directly by existing research on visual organization [3], augmentation and automation [7], and responsible design [5].

Future directions proposed in these works include studying layout interaction patterns [8], comparing spatial vs. linear reading outcomes [3], and designing interventions for mitigating algorithmic bias [2].

By integrating and extending these findings, this concept offers a research-informed, ethically grounded design direction for critical engagement on social media platforms.

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